

PARTS LIST

Semiconductors:

IC1	- AT89C51 microcontroller
IC2	- 74LS04 hex inverter
IC3	- NE555 timer
IC4	- L293D motor driver
IC5	- 7805, 5V regulator
IC6	- 7806, 6V regulator
T1, T2	- BC548 npn transistor
D1-D6	- 1N4007 rectifier diode
LED1-LED5	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 10-kilo-ohm
R2-R5	- 2-kilo-ohm
R6-R10,	
R12, R13	- 220-ohm
R11	- 1-kilo-ohm
VR1, VR2	- 10-kilo-ohm preset

Capacitors:

C1	- 10 μ F, 16V electrolytic
C2, C3	- 33pF ceramic disk
C4, C7, C8	- 0.1 μ F ceramic disk
C5	- 0.01 μ F ceramic disk
C6	- 1000 μ F, 25V electrolytic

Miscellaneous:

X1	- 230V AC primary to 9V, 500mA secondary transformer
X _{AL}	- 12MHz crystal
RL1	- 6V, 1C/O relay
RL2	- 6V, 2C/O relay
M	- 6V DC motor
S1-S9	- Push-to-on switch
LCD module	- 16 $\times$ 2 line

the LCD is initialised and cleared. At power-on reset or manual reset, message "DC motor controller" is displayed on the LCD and LED1 through LED4 turn off. When start switch S1 is pressed, message "Motor Start" is displayed for a second, and then the mode and current speed are displayed. Run LED and clockwise-direction LED are 'on.' Now the program waits for key press. When any key is pressed, the program jumps to one of the functions (start, direction, mode, etc).

Display. It uses the following functions:

1. 'writcmd' function sends command byte to the LCD. It takes one argument byte and sends it to P0.

2. 'writedata' function sends data byte to be displayed on the LCD. It also takes one argument byte and sends it to P0.

3. 'writestr' function writes the whole string (message) on the LCD. It takes pointer as an argument that points address of the first character of the string, then through the pointer it sends all the characters one by one to P0.

4. 'busy' function checks the status of the busy flag of the LCD. If the flag is set, the LCD is not ready and the program remains within the loop. When the flag is reset, the LCD is ready and the program comes out of the loop.

Motor control. It uses the following functions:

1. 'start' function starts motor rotation in one of the modes (continuous, reversible or jogging). The mode is selected by mode-selection flag 'm.' As explained earlier, in continuous mode the motor keeps rotating with the set speed. In reversible mode, the motor changes direction automatically after the set time. In jogging mode, the motor rotates for the set time, stops for a few seconds and then rotates.

2. 'stop' function stops rotating the motor and displays the message on the LCD module. This is indicated by glowing of the stop LED.

3. 'direction' function increments the counter by '1' every time and checks whether the count is even or odd. If the count is even, it selects clock-wise direction, and if the count is odd, it selects anticlockwise direction. This is also indicated on direction CLK and ACLK LEDs.

4. 'mode' function changes modes in cyclic manner. It increases mode-selection flag 'm' every time. If m=0 it selects continuous mode, if m=1 it

selects reversible mode and if m=2 it selects jogging mode. If m=3 it is reset to '0' again and selects continuous mode and likewise.

5. 'incspeed' function increases the width of the pulse by 100 μ s. The generated PWM is of 500 Hz. That means total time is 2 ms=2000 μ s. The width of the pulse is varied from 500 μ s to 1500 μ s in steps of 100 μ s. To display the speed, the variable is first converted into speed factor '1' to '9' and then into ASCII.

6. 'decspeed' function is the same as 'incspeed' but here the width of the pulse is decreased by 100 μ s.

7. 'inctime' function increases the reversible time of the motor by one second. It increases the variable in multiples of 20. To display it on the LCD module, it is first divided by 20 and then converted into ASCII.

8. 'decitime' function is the same as 'inctime' but it decreases reversible time by one second.

Delay. It uses the following functions:

1. 'keyldy' function generates a fix delay of around 50 ms for key debouncing.

2. 'delay' function generates a variable delay by timer 0. The basic delay is of 100 μ s. It is rotated in loop from five to 15 times to generate minimum 500 μ s and maximum 1500 μ s delay.

3. 'time' function again generates a variable delay by timer 1. The basic delay is of 50 ms. It is rotated in multiples of 20 from 20 to 180 to generate minimum 1-second and maximum 9-second delay.

EFY note. The source code and other relevant files of this article are included in this month's EFY-CD.

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DCMC.C

```
#include<reg51.h>
#include<string.h>
sbit rs = P3^1;
// rs pin of LCD
sbit en = P3^0;
// en pin of LCD
```

```
sbit rw = P3^2;
// rw pin of LCD
sbit b = P0^7;
// busy flag
sbit led1=P2^0;
// run indicator
```

```
sbit led2=P2^1;
// stop indicator
sbit led3=P2^2;
// clockwise direction indicator
sbit led4=P2^3;
// anticlockwise direction indicator
```